

Importance Sampling: Subset approach

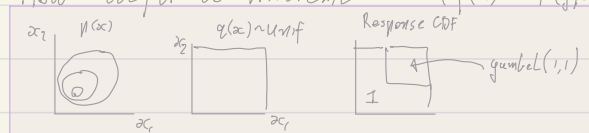
Purpose

Standard importance sampling $E_p[f(x)] = \int f(x) p(x) dx$

assumes $q(x) > 0 \Rightarrow f(x)p(x) > 0$

How useful to Extreme $(f(x) = G(y/x))$

Example:



$\rightarrow$ Can sample rare regions of $p(x)$

Restriction: q need to cover regions where response is guaranteed to be zero, because

$$G(y/x) = \text{Prb}(Y < y/x) = 1 \text{ for } y > 0$$

We then typically choose $q(x)$ so $p(x) \neq 0 \&$
 $\text{supp}(p(x)) \subseteq \text{supp}(q(x))$ [uncommon to find $q(y/x) = 0$]

Improvement: Make an importance

sampling method where $q(x)$ can ignore where response is 0

Benefits:

GP doesn't have to model discontinuity

e.g. ① condition not quantified in
 ② conditions quantified in

Importance sampling more efficient.

q can focus just on regions of non-zero response

In simple terms:

- Want to be able to pick $q(x)$ that just covers regions where response is non-zero
- Current need to ensure $q(x)$ cover all regions where response ≥ 0 & $p(x) > 0$

Variables

$f_x(x)$: pdf of environment data

Y : short term response random variable

$q(y/x)$: pdf of short term response given weather

$G(y/x)$: Cdf of short term response given weather

$h(x)$: importance distribution

$$SF = \text{supp}(f) = \{x \in \mathbb{R}^n : f(x) > 0\}$$

$$SH = \text{supp}(h) = \{x \in \mathbb{R}^n : h(x) > 0\}$$

Assumptions

1) Response is 0 outside of SH

$$h(x) = 0 \Rightarrow Y = 0, G(y/x) = 1$$

Quantity of interest $G(y)$

$$G(y) = \int G(y, x) dx \quad \left[\begin{array}{l} \text{marginalisation} \\ P(A) = \sum P(A, B) f_B \end{array} \right]$$

$$= \int G(y, x) f(x) dx \quad \left[\begin{array}{l} \text{using} \\ P(A, B) = P(A|B)P(B) \end{array} \right]$$

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Calculations over different support

Case: $S_H \subseteq S_F$

$$\begin{aligned} G(y) &= \int G(y|x) f(x) dx \\ &= \int_{S_F} G(y|x) f(x) dx \\ &= \underbrace{\int_{S_H} G(y|x) f(x) dx}_{(1)} + \underbrace{\int_{S_F - S_H} G(y|x) f(x) dx}_{(2)} \end{aligned}$$

$$(1): \int_{S_H} G(y|x) f(x) dx$$

$$= \int_{S_H} G(y|x) \frac{f(x)}{h(x)} h(x) dx$$

$$\approx \frac{1}{N} \sum_{x_i \in h(x)} G(y|x_i) \frac{f(x_i)}{h(x_i)}$$

$\therefore$ (1) can be calculated with standard importance sampling

$$(2) \int_{S_F - S_H} G(y|x) f(x) dx$$

$$= \int_{S_F - S_H} 1 \times f(x) dx \quad [\text{Using Assumption 1}]$$

$$= 1 - \int_{S_H} f(x) dx \quad [\text{using } 1 = P(A) + P(A^c)]$$

$$= 1 - \int_{S_H} \frac{f(x)}{h(x)} h(x) dx \quad [\text{Importance sampling}]$$

$$\approx 1 - \frac{1}{M} \sum_{x_i \in h(x_i)} \frac{f(x_i)}{h(x_i)}$$

$$\therefore G(y) \approx \underbrace{\frac{1}{N} \sum_{x_i \in h(x)} G(y|x_i) \frac{f(x_i)}{h(x_i)}}_{(1)} + \underbrace{1 - \frac{1}{M} \sum_{x_i \in h(x_i)} \frac{f(x_i)}{h(x_i)}}_{(2)}$$

Note: We could use the same samples to estimate both quantities
 $n_{\text{sample}} = \{x_1, \dots, x_n : x_i \in h(x)\}$

$$G(y) \approx \frac{1}{N} \sum_{n_{\text{sample}}} G(y|x_i) \frac{f(x_i)}{h(x_i)} + 1 - \frac{1}{M} \sum_{n_{\text{sample}}} \frac{f(x_i)}{h(x_i)}$$

Note: unbiased but higher variance (i think?)

Case: $S_F \subseteq S_H$

$$G(y) = \int_{S_H} G(y|x) f(x) dx + \int_{S_F - S_H} G(y|x) f(x) dx$$

$$= \int_{S_H} G(y|x) f(x) dx + 0 \quad [S_F - S_H = \emptyset]$$

$$= \int_{S_H} G(y|x) f(x) dx + (1 - 1)$$

$$[\text{using } 1 = \int_{S_H} f(x) dx]$$

$$= \int_{S_H} G(y|x) f(x) dx + (1 - \int_{S_H} f(x) dx)$$

[using importance sampling]

$$= \int_{S_H} G(y|x) \frac{f(x)}{h(x)} h(x) dx + 1 - \int_{S_H} \frac{f(x)}{h(x)} h(x) dx$$

$$\approx \frac{1}{N} \sum_{x_i \in h(x)} G(y|x_i) \frac{f(x_i)}{h(x_i)} + 1 - \frac{1}{M} \sum_{x_i \in h(x_i)} \frac{f(x_i)}{h(x_i)}$$

$\therefore$ for both $S_H \subseteq S_F$ & $S_F \subseteq S_H$ the same formula can be used